

Merchanta AI — Technical Documentation

Version

1.0

Product

Merchanta AI

Document Type

Technical System Documentation

1. Introduction

1.1 Overview

Merchanta AI is an AI-powered e-commerce intelligence platform that analyzes Shopify stores and evaluates how AI systems such as ChatGPT, Gemini, Perplexity, and conversational shopping agents perceive, rank, and recommend products.

The platform enables merchants to optimize products for AI-driven commerce by combining:

- Ecommerce ingestion pipelines
- AI semantic evaluation
- Conversational shopping simulations
- Recommendation readiness scoring
- AI-generated optimization insights

This document explains the technical architecture, backend systems, frontend implementation, AI workflows, infrastructure design, APIs, and deployment architecture.

2. System Architecture Overview

2.1 High-Level Architecture

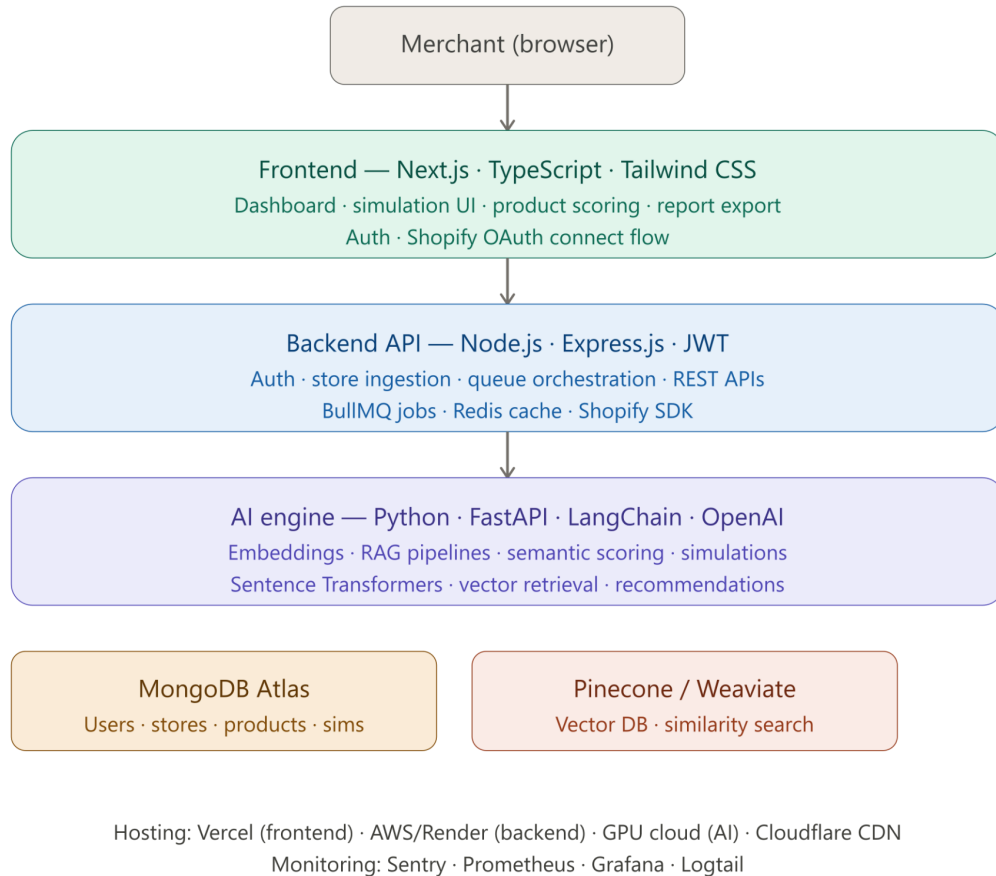


Figure 1—Merchanta AI Technical System Architecture:

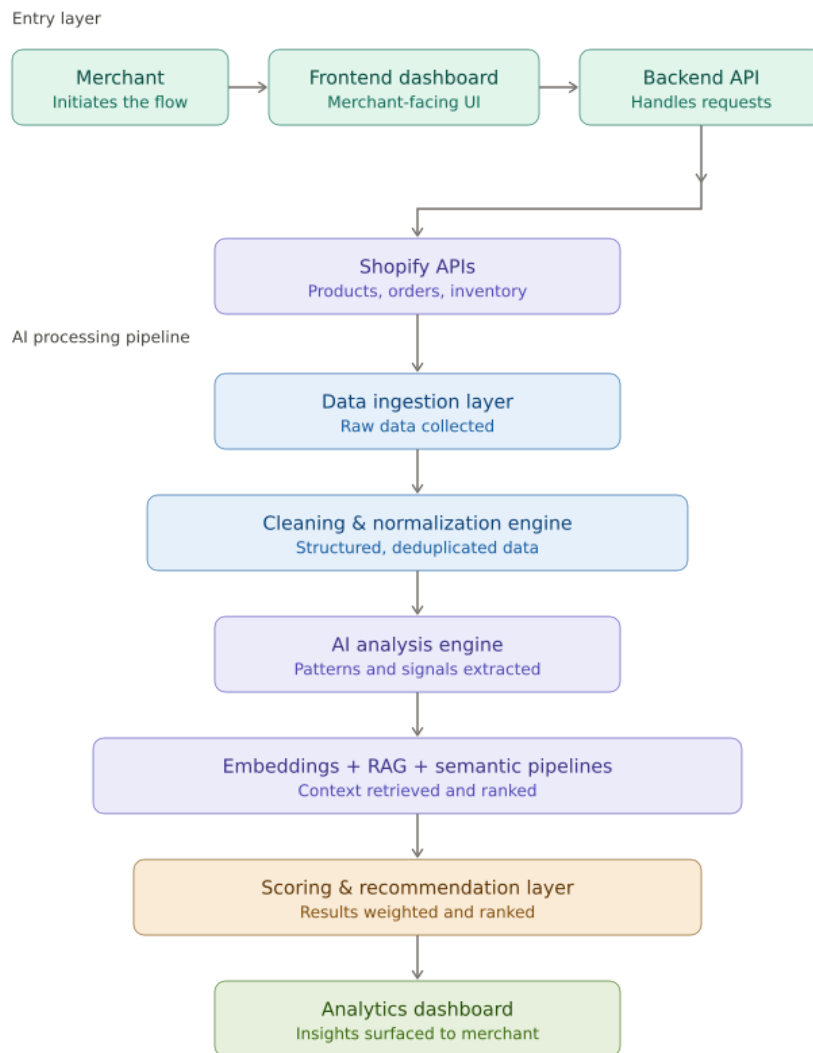
The following architecture illustrates the interaction between frontend services, backend APIs, AI orchestration layers, vector databases and cloud infrastructure.

The platform follows a distributed multi-service architecture.

Core Components

1. Frontend Application
 2. Backend API Server
 3. AI Engine Service
 4. Database Layer
 5. Shopify Integration Layer
 6. Vector Search & Embedding Layer
 7. Analytics & Reporting Layer
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2.2 Architecture Flow



3. Frontend Architecture

3.1 Technology Stack

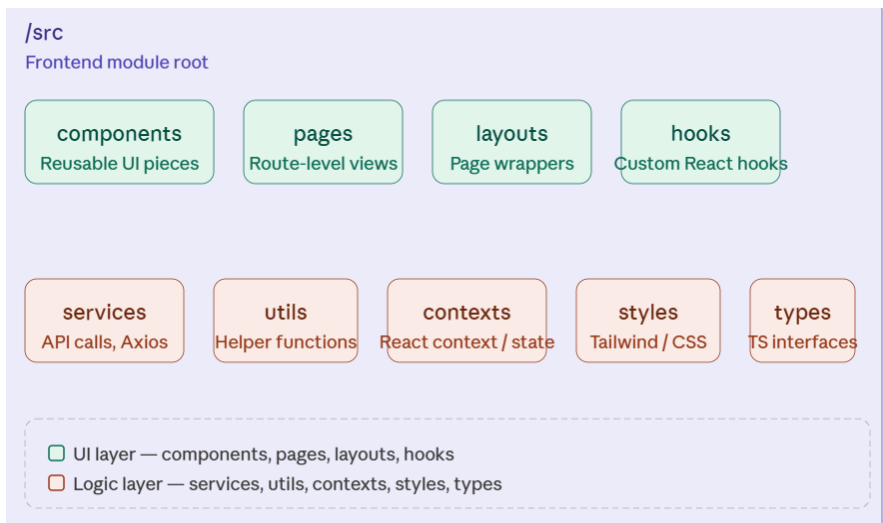
Technology	Purpose
Next.js	Frontend framework
TypeScript	Type-safe frontend development
Tailwind CSS	UI styling
React	Component architecture
Axios	API communication
Chart Libraries	Analytics visualization

3.2 Frontend Responsibilities

The frontend layer is responsible for:

- User authentication
 - Merchant onboarding
 - Shopify connection flow
 - Dashboard rendering
 - Analytics visualization
 - AI simulation interface
 - Product scoring visualization
 - Report generation UI
 - Recommendation display
-

3.3 Frontend Module Structure



3.4 Key UI Components

Dashboard Module

Displays:

- AI visibility scores
- Product analytics
- Recommendation metrics
- Trend charts

Product Analysis Page

Displays:

- Product weaknesses
- AI discoverability metrics
- Semantic analysis
- Optimization recommendations

AI Simulation Interface

Allows merchants to test conversational shopping prompts.

Reporting Module

Generates downloadable reports.

4. Backend Architecture

4.1 Technology Stack

Technology	Purpose
Node.js	Runtime environment
Express.js	API framework
MongoDB	Primary database
Mongoose	ODM layer
JWT	Authentication
Shopify API SDK	Shopify integration

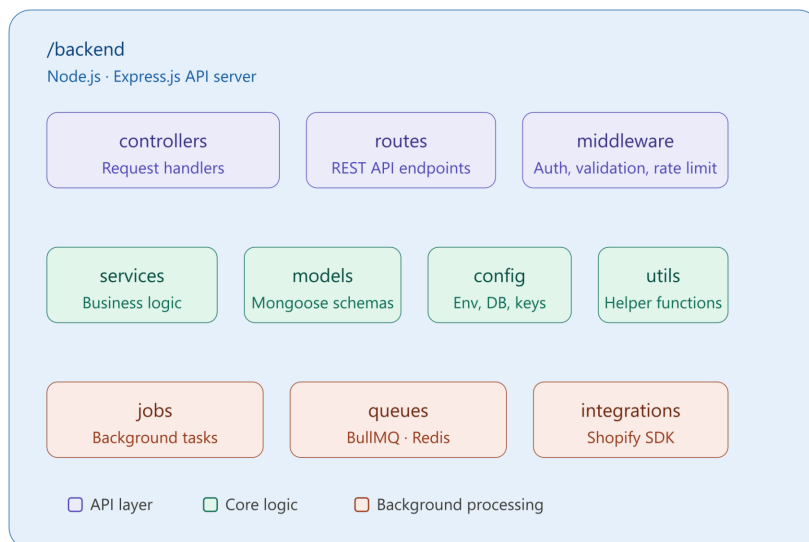
Redis (Optional) Caching layer

4.2 Backend Responsibilities

The backend server manages:

- Authentication
 - Merchant management
 - Shopify store ingestion
 - Product orchestration
 - Queue management
 - API routing
 - Analytics aggregation
 - Report generation
 - AI engine communication
-

4.3 Backend Folder Structure



5. Authentication System

5.1 Authentication Flow

Process

1. Merchant signs up
 2. Backend validates credentials
 3. JWT token generated
 4. Secure session established
 5. Role-based access initialized
-

5.2 Authentication Technologies

Component	Technology
Authentication	JWT
Password Hashing	bcrypt
Session Security	HTTP-only cookies
OAuth Support	Shopify OAuth

6. Shopify Integration Layer

6.1 Purpose

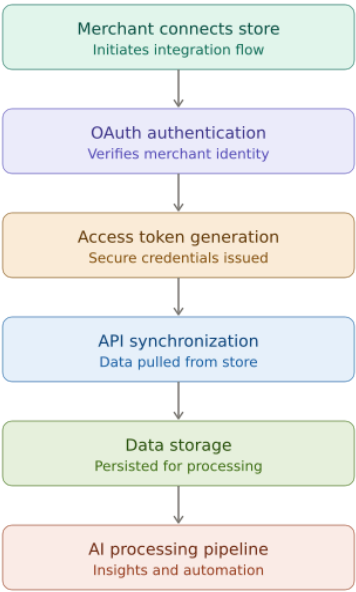
The Shopify integration layer imports merchant e-commerce data into the platform.

6.2 Imported Resources

The platform fetches:

- Products
 - Collections
 - Descriptions
 - Metadata
 - Reviews
 - Policies
 - Tags
 - Inventory data
 - Pricing data
-

6.3 Shopify Integration Workflow



6.4 API Synchronization Strategy

Sync Modes

Mode	Purpose
Initial Sync	Full store ingestion
Incremental Sync	Product updates
Scheduled Sync	Automated refresh

7. Database Design

7.1 Primary Database

MongoDB is used as the primary datastore.

7.2 Core Collections

Users Collection

```
{
  "_id": "ObjectId",
  "name": "string",
  "email": "string",
  "password": "hashed",
  "role": "merchant",
  "createdAt": "date"
}
```

Stores Collection

```
{
  "_id": "ObjectId",
  "userId": "ObjectId",
  "shopifyStoreId": "string",
  "storeName": "string",
  "accessToken": "encrypted",
  "createdAt": "date"
}
```

Products Collection

```
{
  "_id": "ObjectId",
  "storeId": "ObjectId",
  "title": "string",
  "description": "string",
  "metadata": {},
  "collections": [],
  "visibilityScore": "number",
  "trustScore": "number",
  "recommendationReadiness": "number"
}
```

Simulations Collection

```
{
  "_id": "ObjectId",
  "storeId": "ObjectId",
  "query": "string",
  "results": [],
  "confidence": "number",
}
```

```
"createdAt": "date"
}
```

8. AI Engine Architecture

8.1 Technology Stack

Technology	Purpose
Python	AI runtime
FastAPI	AI service framework
LangChain	AI orchestration
OpenAI APIs	LLM processing
Vector Database	Embedding retrieval
Sentence Transformers	Embedding generation

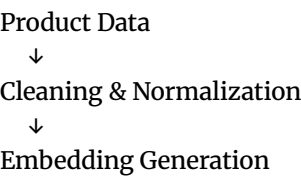
8.2 AI Engine Responsibilities

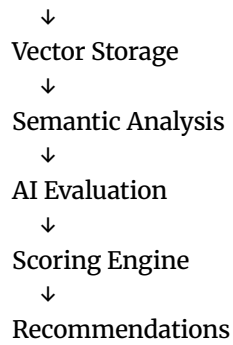
The AI engine performs:

- Semantic analysis
 - Product understanding
 - AI discoverability scoring
 - Conversational simulation
 - Embedding generation
 - Recommendation reasoning
 - RAG retrieval
 - Trust signal evaluation
-

9. AI Processing Pipeline

9.1 Pipeline Overview





9.2 Data Cleaning Pipeline

Tasks Performed

- Remove malformed metadata
 - Normalize formatting
 - Detect missing attributes
 - Standardize descriptions
 - Validate structured data
-

9.3 Embedding Generation

Purpose

Convert product data into vector representations for semantic understanding.

Embedded Data

- Product titles
 - Product descriptions
 - Product metadata
 - Reviews
 - Tags
-

9.4 Vector Database

Responsibilities

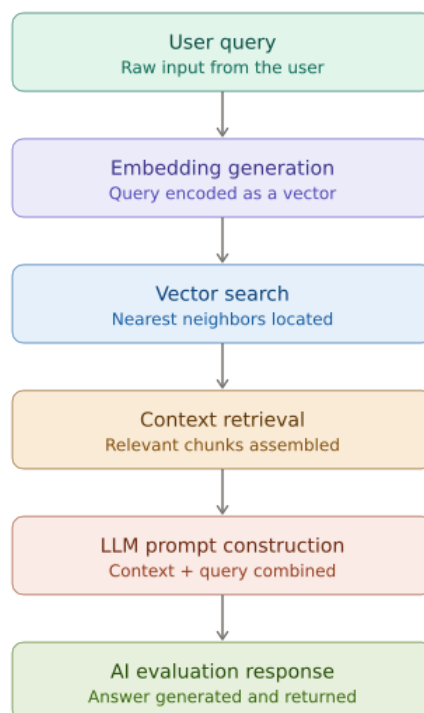
- Semantic retrieval
- Similarity search
- Context retrieval
- Recommendation support

10. RAG Architecture

10.1 Retrieval-Augmented Generation

The platform uses RAG pipelines to provide contextual AI reasoning.

10.2 RAG Workflow



11. AI Visibility Scoring System

11.1 Purpose

Evaluate how effectively AI systems can:

- Understand products
- Recommend products

- Trust product information
 - Compare products
-

11.2 Scoring Dimensions

Metric	Description
Semantic Clarity	Product understanding quality
Trust Signals	Reliability indicators
Metadata Completeness	Structured information quality
Conversational Relevance	AI query compatibility
Recommendation Readiness	AI recommendation suitability

11.3 Score Generation Process



12. AI Shopping Simulation Engine

12.1 Purpose

Simulate real conversational commerce scenarios.

12.2 Simulation Inputs

Example prompts:

- “Best gaming laptop under \$1000”
- “Affordable ergonomic office chair”

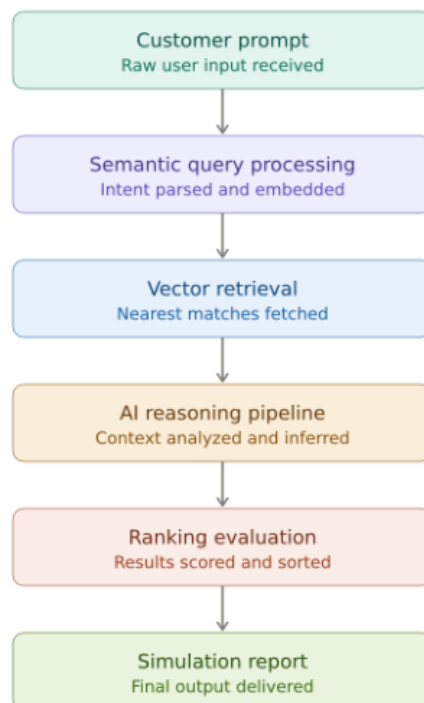
- “Luxury vegan skincare set”
-

12.3 Simulation Outputs

The system evaluates:

- Product ranking probability
 - AI recommendation confidence
 - Semantic match quality
 - Missing optimization areas
-

12.4 Simulation Workflow



13. Recommendation Engine

13.1 Purpose

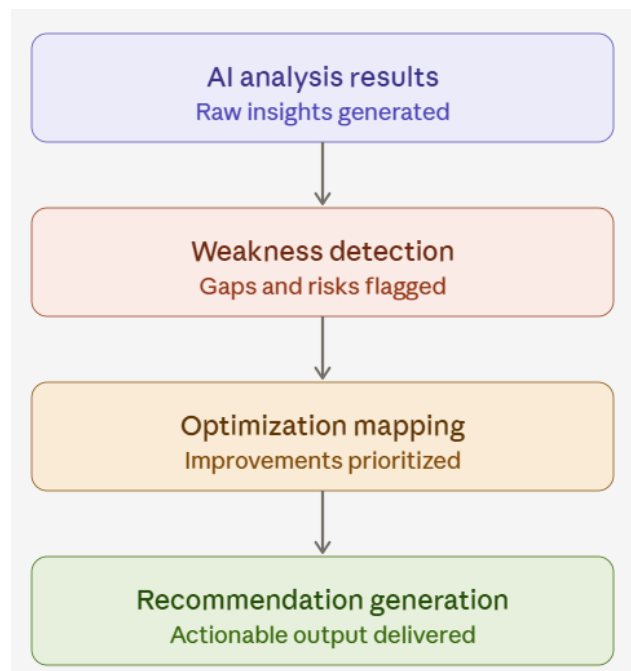
Generate actionable optimization recommendations.

13.2 Recommendation Categories

- Metadata improvements
- Description optimization

- Trust enhancement
 - Semantic clarification
 - Conversational optimization
 - Product comparison readiness
-

13.3 Recommendation Workflow



14. API Design

14.1 REST API Structure

Authentication APIs

POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout

Store APIs

POST /api/store/connect
GET /api/store/details
GET /api/store/products

Analysis APIs

POST /api/analysis/run

GET /api/analysis/results

GET /api/analysis/product/:id

Simulation APIs

POST /api/simulation/run

GET /api/simulation/history

Reporting APIs

GET /api/reports/export

15. Caching Layer

15.1 Redis Usage

Redis can be used for:

- Session caching
 - Queue management
 - API response caching
 - Temporary AI results
 - Rate limiting
-

16. Queue & Background Processing

16.1 Purpose

Background jobs handle long-running AI operations.

16.2 Queue Tasks

- Product ingestion
- Embedding generation
- AI scoring
- Simulation processing

- Report generation
-

16.3 Queue Technologies

Technology	Purpose
BullMQ	Job queues
Redis	Queue backend

17. Security Architecture

17.1 Security Measures

Authentication Security

- JWT authentication
- OAuth validation
- Secure token storage

API Security

- Rate limiting
- Request validation
- Input sanitization

Data Security

- Encrypted access tokens
 - HTTPS communication
 - Secure environment variables
-

18. Scalability Considerations

18.1 Horizontal Scalability

Services are independently scalable:

- Frontend servers
- Backend API servers
- AI engine workers
- Queue workers

- Vector search services

18.2 AI Scalability

AI workloads can be distributed across:

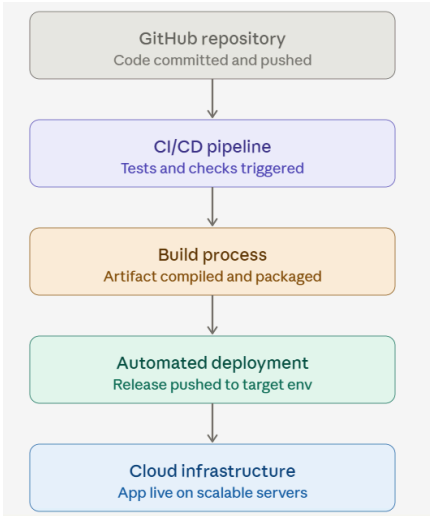
- Multiple FastAPI workers
 - Separate inference servers
 - Dedicated embedding pipelines
-

19. Deployment Architecture

19.1 Deployment Stack

Layer	Technology
Frontend Hosting	Vercel
Backend Hosting	AWS / Render
AI Engine Hosting	GPU-enabled cloud instance
Database	MongoDB Atlas
Vector Database	Pinecone / Weaviate
CDN	Cloudflare

19.2 Deployment Workflow



20. Monitoring & Observability

20.1 Monitoring Components

- API monitoring
 - AI inference tracking
 - Queue monitoring
 - Error logging
 - Usage analytics
-

20.2 Recommended Tools

Tool	Purpose
Sentry	Error tracking
Prometheus	Metrics monitoring
Grafana	Visualization
Logtail	Centralized logging

21. Future Technical Enhancements

Planned Improvements

Multi-Platform Ecommerce Integration

Support:

- WooCommerce
- Magento
- BigCommerce
- Amazon marketplaces

Real-Time Monitoring

Continuous AI ranking analysis.

AI Optimization Automation

Automatic metadata and description generation.

Advanced Recommendation Models

Custom merchant-specific ranking intelligence.

Fine-Tuned Merchant Models

Domain-specific AI optimization pipelines.

22. Conclusion

Merchanta AI is designed as a scalable AI-native ecommerce intelligence platform focused on helping merchants optimize for conversational commerce and AI-driven shopping systems.

The system combines:

- Full-stack web architecture
- AI orchestration pipelines
- Semantic retrieval systems
- Vector-based recommendation analysis
- Conversational simulation frameworks

Through its modular architecture and AI-centric design, Merchanta AI provides merchants with practical intelligence for the next generation of ecommerce discovery and recommendation systems.
